

Abdolamir Karbalaie

Researcher in Trustworthy AI & Machine Learning

Stockholm, Sweden | September 2026

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Research profile

I develop trustworthy, reproducible, and uncertainty-aware machine learning for dependent and multimodal data. My research connects medical imaging, biomechanics, clinical speech, computer vision, and critical infrastructure through a common focus on rigorous validation, explainability, and human oversight. As lead developer of TrustCV, I turn evaluation methods into reusable scientific software and work with clinicians, engineers, and researchers to connect analysis with practical questions.

Current appointments

Affiliated Researcher & Lead Developer

Karolinska Institutet - CLINTEC / SMAILE Core Facility | 2025 - Present

- Lead developer of TrustCV: validation architecture, leakage checks, reusable Python tooling, and methodological documentation.
- Co-developed MEDLEY-BENCH; contributed formal analysis, human validation, and benchmark interpretation.
- Co-develop TRIAGE-MEDLEY: transcript alignment, model disagreement, uncertainty surfacing, and human-review workflows.
- Corresponding author and co-supervisor of an international scoping review of AI in nailfold capillaroscopy.

Machine Learning Lead - Railway Climate Resilience

Luleå University of Technology | 2023 - Present

- Lead machine-learning modelling for prediction and classification of climate-related railway failures.
- Develop reproducible evaluation workflows with MLflow and Docker, and interpret model behaviour with infrastructure specialists.
- First author of the 2025 Climate Risk Management study on regional and national railway networks.

Research experience

Staff Scientist, AI & Biomechanics

Umeå University | Jan 2025 - Aug 2025

- Developed ML models for movement-analysis and rehabilitation data using Python, PyTorch/scikit-learn, and SHAP.
- Owned data preparation, model development, evaluation, experiment tracking, and containerised deployment in a research setting.
- Worked with clinicians and engineers to make research outputs interpretable and usable.

Postdoctoral Researcher - ML for Biomechanics

Umeå University | 2023 - 2025

- Developed multimodal models connecting biomechanical motion and EMG measurements with fear of re-injury after ACL reconstruction.
- Led study design, participant-aware validation, multimodal analysis, interpretation, and first-author manuscripts.
- Co-supervised student research and contributed reusable workflows for repeated-measures data.

Postdoctoral Researcher - Video Analytics & XAI

Mid Sweden University | 2019 - 2021

- Developed and trained deep learning models (CNNs/RNNs) for anomaly and event detection in real-time surveillance video streams.
- Optimised models for latency and throughput; implemented streaming pipelines, monitoring, and robust evaluation under distribution shift.
- Contributed to both algorithm development and software implementation in a small research/engineering team.

Doctoral Researcher - Medical Image Analysis

KTH Royal Institute of Technology | 2013 - 2019

- Built computer-aided diagnostic methods for rheumatic disease based on nailfold capillaroscopy images (microvascular medical imaging).
- Developed image-processing and ML algorithms for capillary detection, quantification, and feature extraction to support clinical decision-making.
- Worked with rheumatologists and imaging units to design analysis pipelines, validate models, and prepare scientific and technical documentation.

Education & technical practice

Ph.D., Applied Medical Technology

KTH Royal Institute of Technology | 2018

Thesis: Novel Analysis Toolkit for Capillaroscopic Images: Development and Clinical Evaluation. Research in medical image analysis, quantitative validation, and clinical decision-support technology.

M.Sc., Mathematics

Isfahan University of Technology | 2002

Advanced mathematical training that supports my work in modelling, statistical reasoning, and scientific computation.

B.Sc., Mathematics

University of Isfahan | 2000

Foundations in mathematics, developed into a research practice spanning computational methods and applied machine learning.

Technical practice

Languages - Python (primary), R, MATLAB, SQL, Bash

ML / AI - PyTorch, TensorFlow, scikit-learn, MONAI, Tree ensembles & boosting, Deep learning for image & video

Automation - n8n, RAG pipelines, ETL pipelines, API integration

MLOps - MLflow, Docker, Kubernetes, Experiment tracking, Models as REST services, Configuration-driven pipelines

Evaluation - Cross-validation design, Leakage detection, Calibration, Metric selection, MedTech validation documentation

Tools - Git/GitHub, Linux, Jupyter/Colab, Azure/AWS (basic)

Languages

English (Fluent) | Swedish (Intermediate) | Persian (Native) | Arabic (Native)

Teaching & academic service

Teaching & supervision

- Probability & Statistics, Numerical Analysis, Calculus I-III, Data Mining (SAS)
- Teaching assistant: 3D Image Reconstruction in Medicine (KTH)
- Supervision of MSc and BSc projects in ML/imaging; current PhD co-supervision in medical imaging/XAI
- Pedagogy: UPL - Education for Sustainable Development
- Teaching approach: connect mathematical foundations to applied problems, reproducible ML practice, leakage prevention, and transparent reporting.

Leadership & service

- Lead developer and maintainer of TrustCV, coordinating requirements from multiple research groups and ensuring long-term maintainability.
- Experience in small, cross-functional teams with clinicians, engineers, and researchers in Sweden and internationally.
- Active reviewer for Scientific Reports, Arthritis Research & Therapy, and Signal, Image and Video Processing.
- Corresponding author and co-supervisor of an AI-in-capillaroscopy review spanning Karolinska Institutet and collaborating universities in Iran.
- Co-developer of MEDLEY-BENCH, recognised with a Kaggle Grand Prize in 2026.

Research focus

I connect methodological research with reusable software: participant-aware evaluation, uncertainty in clinical speech, multimodal rehabilitation analysis, climate-related infrastructure modelling, and quantitative medical imaging.

Selected publications

Nine selected works. Publication status is stated for each entry; preprints have not completed peer review.

1. Artificial Intelligence in Nailfold Capillaroscopy: A Scoping Review of Validation, Reproducibility, and Clinical Translation

Emrani, Z., Setarehdan, S. K., Jafari-Varzaneh, H., Fatemi, A., & Karbalaie, A.

Research Square (preprint; manuscript under review) | 2026 | Preprint

<https://doi.org/10.21203/rs.3.rs-10348706/v1>

2. Cross-model disagreement as a reference-free signal for prioritizing human review in medical speech transcription

Karbalaie, A., Seoane, F., & Abtahi, F.

Frontiers in Artificial Intelligence | 2026 | Journal

<https://doi.org/10.3389/frai.2026.1829902>

3. Enhancing fear of re-injury classification after ACL reconstruction by integrating biomechanical and electromyography data using multimodal machine learning methods

Karbalaie, A., Grinberg, A., Strong, A., Grip, H., Prorok, K., Häger, C. K., & Nordström, T.

Journal of Biomechanics | 2026 | Journal

<https://doi.org/10.1016/j.jbiomech.2026.113346>

4. Participant-Aware Model Validation for Repeated-Measures Data: Comparative Cross-Validation Study

Karbalaie, A., Abtahi, F., & Häger, C. K.

JMIR AI | 2026 | Journal

<https://doi.org/10.2196/87728>

5. Cross-validation under data dependence: a review-derived taxonomy and TrustCV, a leakage-aware Python toolkit

Karbalaie, A. & Abtahi, F.

Research Square (preprint) | 2026 | Preprint

<https://doi.org/10.21203/rs.3.rs-9357577/v1>

6. MEDLEY-BENCH: Benchmarking Behavioural Metacognition and Belief Revision Under Social Pressure in Large Language Models

Abtahi, F., Karbalaie, A., Illueca-Fernandez, E., & Seoane, F.

arXiv (preprint) | 2026 | Preprint

<https://arxiv.org/abs/2604.16009v2>

Selected publications / continued

7. Beyond self-reports after anterior cruciate ligament injury - machine learning methods for classifying and identifying movement patterns related to fear of re-injury

Karbalaie, A., Strong, A., Nordström, T., Schelin, L., Selling, J., Grip, H., Prorok, K., & Häger, C. K.

Journal of Sports Sciences | 2026 | Journal

<https://doi.org/10.1080/02640414.2025.2578584>

8. Classifying climate-related failures for regional-national railway networks

Karbalaie, A., Soleimani-Chamkhorami, K., Famurewa, S. M., & Garmabaki, A. H. S.

Climate Risk Management | 2025 | Journal

<https://doi.org/10.1016/j.crm.2025.100764>

9. Identifying climate-related failures in railway infrastructure using machine learning

Soleimani-Chamkhorami, K., Karbalaie, A., Kasraei, A., Haghighi, E., Famurewa, S. M., & Garmabaki, A. H. S.

Transportation Research Part D: Transport and Environment | 2024 | Journal

<https://doi.org/10.1016/j.trd.2024.104371>